
Online Examination Report

Date:	21.07.2014
Attempts:	2
Examinee:	Ariho, William
Date of birth:	4/22/1989
Subject:	030-Flight Performance&Planing
Result:	31,7 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	388	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	374	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
3	420	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
4	377	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
5	379	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
6	376	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
7	378	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
8	387	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
9	381	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
10	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	315	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
12	390	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
13	391	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
14	183	31.2	LOADING	0,00	1,00
15	278	31.3.3	Basic calculations of CG	0,00	2,00
16	425	32.	PERFORMANCE - AEROPLANES	0,00	1,00
17	423	32.	PERFORMANCE - AEROPLANES	0,00	1,00
18	339	32.	PERFORMANCE - AEROPLANES	0,00	1,00
19	410	32.	PERFORMANCE - AEROPLANES	0,00	1,00
20	61	32.2.5.1	Take-off	1,00	1,00
21	75	32.2.5.4	Landing	1,00	1,00
22	77	32.2.5.4	Landing	1,00	1,00
23	132	32.3.3.1	Take-off	0,00	1,00
24	210	32.3.3.1	Take-off	0,00	1,00
25	95	32.4.1.1	Definitions of terms used	0,00	1,00
26	135	32.4.1.7	Take-off climb	0,00	1,00
27	29	32.4.1.8	Obstacle-limited take-off	0,00	1,00
28	27	32.4.3	Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	68	32.4.3	Cruise	0,00	1,00
30	21	32.4.5.1	Descent techniques	1,00	1,00
31	414	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
32	415	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
33	18	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
34	92	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
35	419	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	1,00	1,00
36	417	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	1,00	1,00
37	226	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
38	148	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
39	2	33.2.1	IFR Navigation plan	0,00	1,00
40	122	33.3.3	Specific fuel calculation procedures	1,00	1,00
Total:32%				13,00	41,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Ariho, William

Location: Uganda Civil Aviation Authority

Your Result: 13,00 from 41,00 Points (31,71 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	3,00	16,00	18,75
31.2. LOADING	1	0,00	1,00	0,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	1	0,00	2,00	0,00
32. PERFORMANCE - AEROPLANES	15	4,00	15,00	26,67
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	3	3,00	3,00	100,00
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	2	0,00	2,00	0,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	6	1,00	6,00	16,67
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	6,00	10,00	60,00
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	4	2,00	4,00	50,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	0,00	1,00	0,00
33.3. FUEL PLANNING	1	1,00	1,00	100,00
Total	40	13,00	41,00	31,71

1

What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?

- ☐ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.
- ☐ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.
- ☒ There is no relationship between CG location and indicated airspeed.

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2

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb
CG location Station 90.0
Aft CG limit Station 90.5

☐ 61.0 pounds.

☐ 161.0 pounds.

☒ 110.5 pounds.

3

The center of gravity tube is

- ☐ lengthened for lighter pilots.
- ☒ shortened for lighter pilots.
- ☐ lengthened for heavier pilots.

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4

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 81.3.

☒ 83.4.

☐ 80.6.

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5

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☒ 68.9 pounds.

☐ 62.5 pounds.

☐ 65.8 pounds.

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6

What is the location of the CG if 90 pounds are removed from Station 140?

Aircraft weight 6,230 lb

CG location Station 79

☒ **79.9.**

☐ **77.9.**

☐ **78.1.**

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7

What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☐ 84.9.

☐ 84.1.

☒ 85.1.

8

As the CG moves aft, an aircraft becomes

- ☐ less stable, yet easier to control.
- ☒ less stable and less controllable.
- ☐ more stable and controllable as long as the aft CG is not exceeded.

9

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

- ☒ Yes; the CG would remain at Station 115.8.
- ☐ Yes; the new CG would be located at Station 117.69.
- ☐ No; the new CG would be located at Station 118.15.

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10

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☐ 924.5 pounds.

☒ 953.6 pounds.

☐ 968.6 pounds.

11

The term 'weigh-off' as used in ballooning means to determine the

- ☐ amount of fuel required for ascent to a preselected altitude.
- ☐ static equilibrium of the balloon as loaded for flight.
- ☒ standard weight and balance of the balloon.

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12

Under which condition is a forward CG most critical?

☐ When in an unusual attitude.

☐ On landing.

☒ On takeoff.

13

As the CG location is changed, recovery from a stall becomes progressively

- ☐ less difficult as the CG moves rearward.
- ☐ more difficult as the CG moves either forward or rearward.
- ☒ more difficult as the CG moves rearward.

14

In flight the wing of an aircraft containing fuel is subjected to vertical loads that produce a bending moment which is:

- ☒ highest at the wing root
- ☐ equal to half the weight of the aircraft multiplied by the semi span
- ☒ equal to the zero -fuel weight multiplied by the span
- ☐ lowest at the wing root

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15

(Please view annex 031-038 rev. 05.11.2004)
For the following boom to be in balance:

See Attachments:

1. 031-038.gif

Annex

(Attachment
No.1)

☒ $A = B \times Fb / Fa$

☐ $A = B \times Fa / Fb$

☒ $A = B (Fa + Fb)$

☐ $A = B + Fb / Fa$

16

During recoveries from unusual attitudes, level flight is attained the moment

- ☒ **a zero rate of climb is indicated on the VSI.**
- ☐ **the horizon bar on the attitude indicator is exactly overlapped with the miniature airplane.**
- ☐ **the altimeter and airspeed needles stop prior to reversing their direction of movement.**

17

To level off at an airspeed higher than the descent speed, the addition of power should be made, assuming a 500 FPM rate of descent, at approximately

- ☐ 150 to 200 feet above the desired altitude.
- ☐ 100 to 150 feet above the desired altitude.
- ☒ 50 to 100 feet above the desired altitude.

18

(Refer to Figure 89.) Which of the following configurations will result in the shortest landing distance over a 50-foot obstacle to a wet runway?

See Attachments:

1. atp_89.jpg

figure 89

(Attachment
No.2)

- ☒ Brakes and spoilers at 122,500 pounds gross weight.
- ☐ Brakes, spoilers, and reversers at 131,000 pounds gross weight.
- ☐ Brakes and reversers at 124,000 pounds gross weight.

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19

At what minimum speed will dynamic hydroplaning begin if a tire has an air pressure of 70 PSI?

☐ 75 knots.

☐ 80 knots.

☒ 85 knots.

20

What effect has a downhill slope on the take-off speeds?The slope

- ☒ **decreases the take-off speed V1.**
- ☐ **increases the IAS for take-off.**
- ☐ **decreases the TAS for take-off.**
- ☐ **has no effect on the take-off speed V1.**

21

For a turboprop powered aeroplane, a 2200 m long runway at the destination aerodrome is expected to be "wet". The "dry runway" landing distance, should not exceed:

☐ 1540 m.

☒ 1339 m.

☐ 1771 m.

☐ 1147 m.

22

A runway is contaminated with 0.5 cm of wet snow. The flight manual of a light twin nevertheless authorises a landing in these conditions. The landing distance will be, in relation to that for a dry runway:

☐ unchanged

☐ reduced

☒ increased

☐ substantially decreased

23

On a dry runway the accelerate stop distance is increased

- ☐ by headwind.
- ☒ by a lower take-off mass because the aeroplane accelerates faster to V1.
- ☒ by uphill slope.
- ☐ by low outside air temperature.

24

Due to standing water on the runway the field length limited take-off mass will be

☐ unaffected.

☐ only higher for three and four engine aeroplanes.

☒ higher.

☐ lower.

25

The decision speed at take-off (V1) is the calibrated airspeed:

- ☐ *below which take-off must be rejected if an engine failure is recognized, above which take-off must be continued.*
- ☒ *below which the take-off must be continued.*
- ☐ *at which the failure of the critical engine is expected to occur.*
- ☐ *at which the take-off must be rejected.*

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26

With an true airspeed of 194 kt and a vertical speed of 1 000 ft/min, the climb gradient is about :

☐ 0.03

☐ 0.08

☐ 3°

☒ 5°

27

Which statement, in relation to the climb limited take-off mass of a jet aeroplane, is correct?

- ☐ On high elevation airports equipped with long runways the aeroplane will always be climb limited.
- ☒ The climb limited take-off mass is determined at the speed for best rate of climb.
- ☐ The climb limited take-off mass decreases with increasing OAT.
- ☐ 50% of a head wind is taken into account when determining the climb limited take-off mass.

28

Under which condition should you fly considerably lower (4 000 ft or more) than the optimum altitude ?

- ☒ **If the maximum altitude is below the optimum altitude.**
- ☐ **If at the lower altitude either considerably less headwind or considerably more tailwind can be expected.**
- ☐ **If at the lower altitude either more headwind or less tailwind can be expected.**
- ☐ **If the temperature is lower at the low altitude (high altitude inversion).**

29

At constant thrust and constant altitude the fuel flow of a jet engine

- ☐ is independent of the airspeed.
- ☐ increases with decreasing OAT.
- ☒ increases slightly with increasing airspeed.
- ☒ decreases slightly with increasing airspeed.

30

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ A tailwind component decreases the ground distance.
- ☐ A headwind component increases the ground distance.
- ☐ A tailwind component increases fuel and time to descent.
- ☒ A tailwind component increases the ground distance.

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31

GIVEN:

Usable fuel at takeoff 40 gal
Fuel consumption rate 12.2 gal/hr
Constant groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 215 NM.

☐ 121 NM.

☒ 176 NM.

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32

If fuel consumption is 15.3 gallons per hour and groundspeed is 167 knots, how much fuel is required for an aircraft to travel 620 NM?

☒ 57 gallons.

☐ 60 gallons.

☐ 63 gallons.

33

Which of the following statements regarding filing a flight plan is correct?

- ☐ Any flight plan should be filed at least 10 minutes before departure.
- ☐ A flying college can file repetitive flight plan for VFR flights.
- ☒ A flight plan should be filled atleast one hour before departure
- ☐ A flight plan should be filed when a national FIR boundary will be crossed.

34

From which of the following would you expect to find details of the Search and Rescue organisation and procedures (SAR) ?

☐ SIGMET

☒ AIP (Aeronautical Information Publication)

☒ NOTAM

☐ ATCC broadcasts

35

All fuel tanks should be fired during preflight to determine

- ☐ if there are any leaks in the tanks.
- ☐ if the pilot light functions properly on each tank.
- ☒ burner pressure and condition of the valves.

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36

How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?

☒ 1 hour.

☐ 30 minutes.

☐ 2 hours.

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37

Given : true track 017; W/V 340/30; TAS 420 kt Find : wind correction angle (WCA) and ground speed (GS)

☒ WCA -2° ; GS 426 kt

☐ WCA -2° ; GS 396 kt

☐ WCA $+2^{\circ}$; GS 396 kt

☐ WCA $+2^{\circ}$; GS 416 kt

38

From which of the following would you expect to find information regarding known short unserviceability of VOR, TACAN, and NDB ?

☐ D ATCC broadcasts

☒ A NOTAM

☐ C SIGMET

☒ B AIP (Aeronautical Information Publication)

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39

In an ATS flight plan, Item 15 (route), a cruising pressure altitude of 32000 feet would be entered as :

☒ A FL320

☐ C 32000

☐ D F320

☐ B S3200

40

An aircraft flies at an airspeed of 380 kt. It flies from A to B and back to A. Distance AB = 480 NM. When going from A to B, it experiences a headwind component = 60 kt. The wind remains constant. The duration of the flight will be:

☐ 2h 32min

☐ 3h 00min

☐ 2h 10min

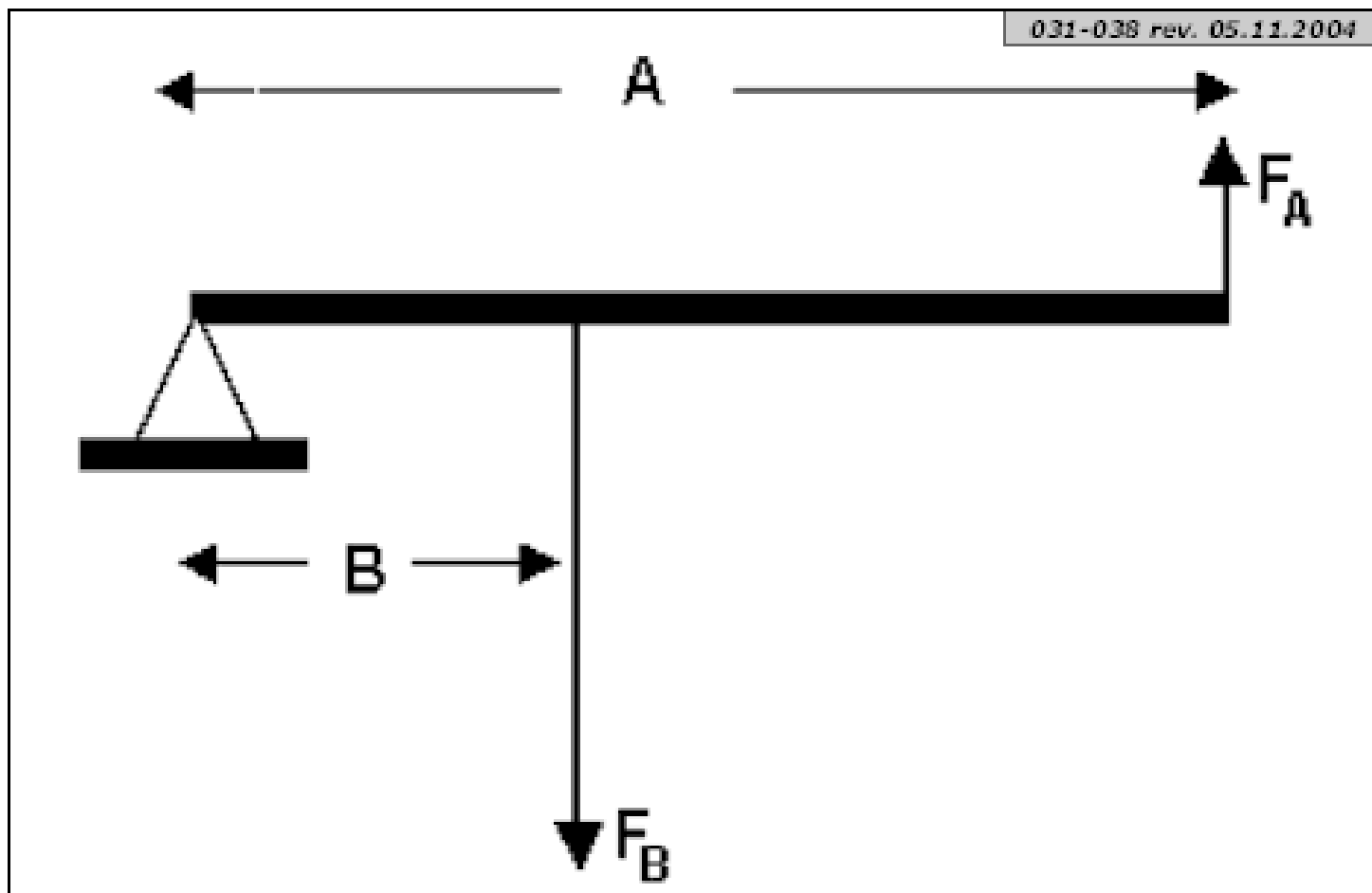
☒ 2h 35min

Attachment No. 1

Questions 15

Annex

031-038.gif

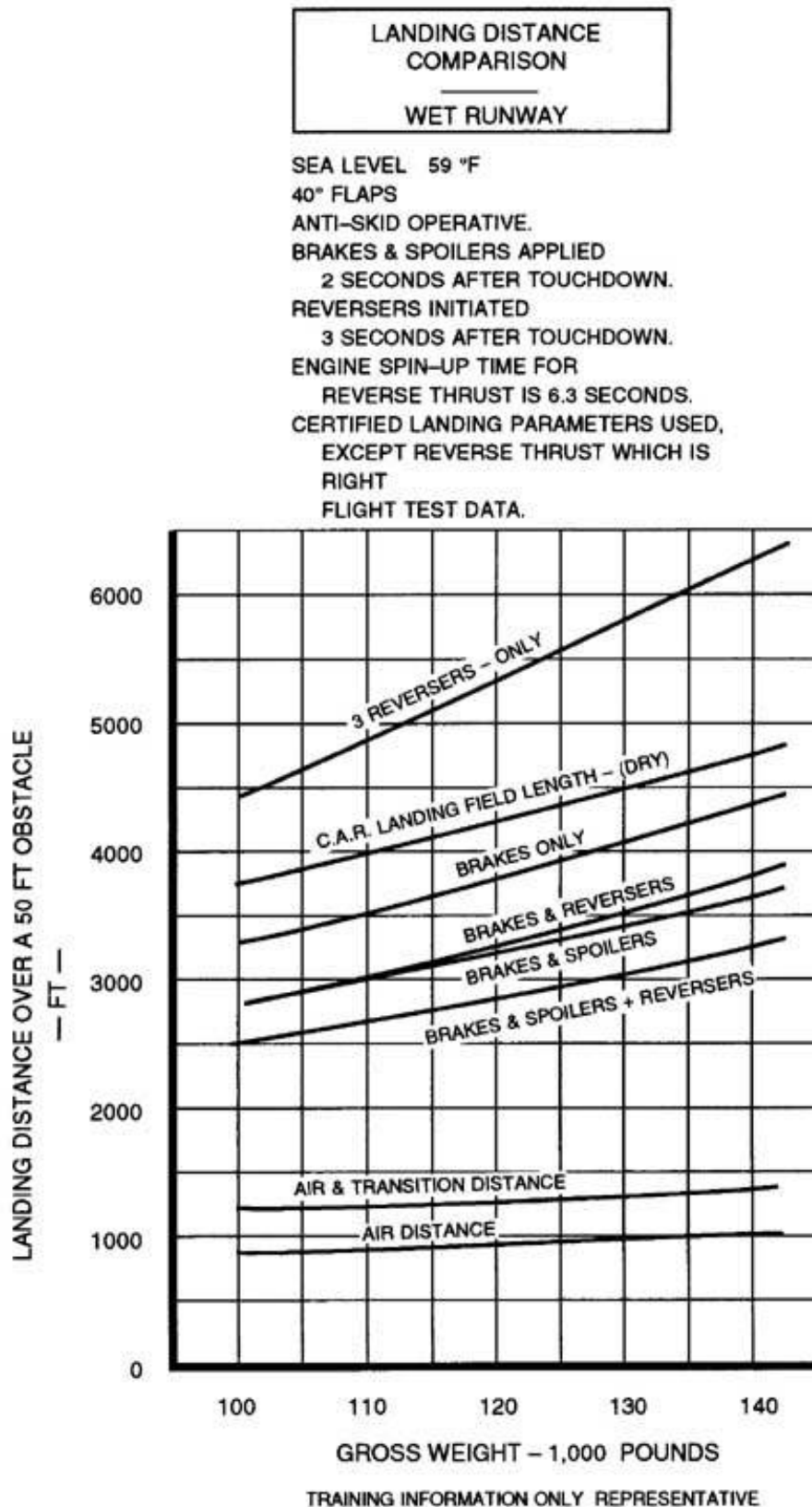


Attachment No. 2

Questions 18

figure 89

atp_89.jpg



NORMAL LANDING

Figure 89. B-727 – Normal Landing – Wet Runway © ASA